

Class 11: AlphaFold

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```
library(bio3d)

pdb <- read.pdb("hivpr_dimer_23119/hivpr_dimer_23119_unrelaxed_rank_002_alphafold2_multimer_v1.pdb")

pdb
```

Call: read.pdb(file = "hivpr_dimer_23119/hivpr_dimer_23119_unrelaxed_rank_002_alphafold2_multimer_v1.pdb")

```
Total Models#: 1
Total Atoms#: 1514, XYZs#: 4542 Chains#: 2 (values: A B)

Protein Atoms#: 1514 (residues/Calpha atoms#: 198)
Nucleic acid Atoms#: 0 (residues/phosphate atoms#: 0)

Non-protein/nucleic Atoms#: 0 (residues: 0)
Non-protein/nucleic resid values: [ none ]
```

```
Protein sequence:
PQITLWQRPLVTIKIGGQLKEALLDTGADDTVLEEMSLPGRWKPKMIGGIGGFIKVRQYD
QILIEICGHKAIGTVLVGPTPVNIIGRNLLTQIGCTLNFPQITLWQRPLVTIKIGGQLKE
ALLDTGADDTVLEEMSLPGRWKPKMIGGIGGFIKVRQYDQILIEICGHKAIGTVLVGPTP
VNIIGRNLLTQIGCTLNF
```

+ attr: atom, xyz, calpha, call

Make a vector of input PDB file names that we can read into R

```
pdbfiles <- list.files("hivpr_dimer_23119/",
                       pattern = ".pdb",
                       full.names = TRUE)
```

```
pdb <- read.pdb("1hsg")
```

Note: Accessing on-line PDB file

```
pdbbs <- pdbaln(pdbfiles, fit = TRUE, exefile = "msa")
```

Reading PDB files:

```
hivpr_dimer_23119//hivpr_dimer_23119_unrelaxed_rank_001_alphafold2_multimer_v3_model_4_seed_0
hivpr_dimer_23119//hivpr_dimer_23119_unrelaxed_rank_002_alphafold2_multimer_v3_model_1_seed_0
hivpr_dimer_23119//hivpr_dimer_23119_unrelaxed_rank_003_alphafold2_multimer_v3_model_5_seed_0
hivpr_dimer_23119//hivpr_dimer_23119_unrelaxed_rank_004_alphafold2_multimer_v3_model_2_seed_0
hivpr_dimer_23119//hivpr_dimer_23119_unrelaxed_rank_005_alphafold2_multimer_v3_model_3_seed_0
.....
```

Extracting sequences

```
pdb/seq: 1   name: hivpr_dimer_23119//hivpr_dimer_23119_unrelaxed_rank_001_alphafold2_multimer_v3_model_4_seed_0
pdb/seq: 2   name: hivpr_dimer_23119//hivpr_dimer_23119_unrelaxed_rank_002_alphafold2_multimer_v3_model_1_seed_0
pdb/seq: 3   name: hivpr_dimer_23119//hivpr_dimer_23119_unrelaxed_rank_003_alphafold2_multimer_v3_model_5_seed_0
pdb/seq: 4   name: hivpr_dimer_23119//hivpr_dimer_23119_unrelaxed_rank_004_alphafold2_multimer_v3_model_2_seed_0
pdb/seq: 5   name: hivpr_dimer_23119//hivpr_dimer_23119_unrelaxed_rank_005_alphafold2_multimer_v3_model_3_seed_0
```

```
pdbbs
```

```

1               .               .               .               .               50
[Truncated_Name:1]hivpr_dime PQITLWQRPLVTIKIGGQLKEALLDTGADDTVLEEMSLPGRWKPKMIGGI
[Truncated_Name:2]hivpr_dime PQITLWQRPLVTIKIGGQLKEALLDTGADDTVLEEMSLPGRWKPKMIGGI
[Truncated_Name:3]hivpr_dime PQITLWQRPLVTIKIGGQLKEALLDTGADDTVLEEMSLPGRWKPKMIGGI
[Truncated_Name:4]hivpr_dime PQITLWQRPLVTIKIGGQLKEALLDTGADDTVLEEMSLPGRWKPKMIGGI
[Truncated_Name:5]hivpr_dime PQITLWQRPLVTIKIGGQLKEALLDTGADDTVLEEMSLPGRWKPKMIGGI
*****
1               .               .               .               .               50

51             .               .               .               .               100
[Truncated_Name:1]hivpr_dime GGFIKVRQYDQILIEICGHKAIGTVLVGPTPVNIIGRNLLTQIGCTLNFP
[Truncated_Name:2]hivpr_dime GGFIKVRQYDQILIEICGHKAIGTVLVGPTPVNIIGRNLLTQIGCTLNFP
[Truncated_Name:3]hivpr_dime GGFIKVRQYDQILIEICGHKAIGTVLVGPTPVNIIGRNLLTQIGCTLNFP
[Truncated_Name:4]hivpr_dime GGFIKVRQYDQILIEICGHKAIGTVLVGPTPVNIIGRNLLTQIGCTLNFP
[Truncated_Name:5]hivpr_dime GGFIKVRQYDQILIEICGHKAIGTVLVGPTPVNIIGRNLLTQIGCTLNFP
*****
```

```

51          .          .          .          .          100

101         .          .          .          .          150
[Truncated_Name:1]hivpr_dime  QITLWQRPLVTIKIGGQLKEALLDTGADDTVLEEMSLPGRWPKMIGGIG
[Truncated_Name:2]hivpr_dime  QITLWQRPLVTIKIGGQLKEALLDTGADDTVLEEMSLPGRWPKMIGGIG
[Truncated_Name:3]hivpr_dime  QITLWQRPLVTIKIGGQLKEALLDTGADDTVLEEMSLPGRWPKMIGGIG
[Truncated_Name:4]hivpr_dime  QITLWQRPLVTIKIGGQLKEALLDTGADDTVLEEMSLPGRWPKMIGGIG
[Truncated_Name:5]hivpr_dime  QITLWQRPLVTIKIGGQLKEALLDTGADDTVLEEMSLPGRWPKMIGGIG
*****
101         .          .          .          .          150

151         .          .          .          .          198
[Truncated_Name:1]hivpr_dime  GFIKVRQYDQILIEICGHKAIGTVLVGPTPVNIIGRNLLTQIGCTLNF
[Truncated_Name:2]hivpr_dime  GFIKVRQYDQILIEICGHKAIGTVLVGPTPVNIIGRNLLTQIGCTLNF
[Truncated_Name:3]hivpr_dime  GFIKVRQYDQILIEICGHKAIGTVLVGPTPVNIIGRNLLTQIGCTLNF
[Truncated_Name:4]hivpr_dime  GFIKVRQYDQILIEICGHKAIGTVLVGPTPVNIIGRNLLTQIGCTLNF
[Truncated_Name:5]hivpr_dime  GFIKVRQYDQILIEICGHKAIGTVLVGPTPVNIIGRNLLTQIGCTLNF
*****
151         .          .          .          .          198

```

Call:

```
pdbaln(files = pdbfiles, fit = TRUE, exefile = "msa")
```

Class:

```
pdb, fasta
```

Alignment dimensions:

```
5 sequence rows; 198 position columns (198 non-gap, 0 gap)
```

```
+ attr: xyz, resno, b, chain, id, ali, resid, sse, call
```

```
rd <- rmsd(pdb, fit = TRUE)
```

Warning in rmsd(pdb, fit = TRUE): No indices provided, using the 198 non NA positions

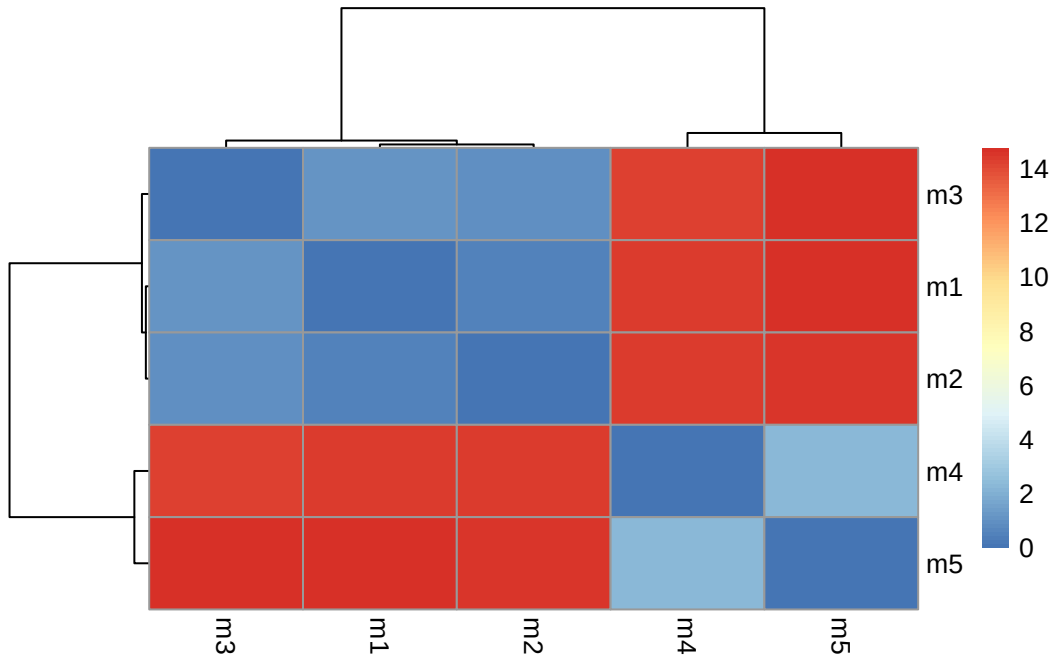
```
range(rd)
```

```
[1] 0.000 14.754
```

```
library(pheatmap)

colnames(rd) <- paste0("m", 1:5)
rownames(rd) <- paste0("m", 1:5)

pheatmap(rd)
```

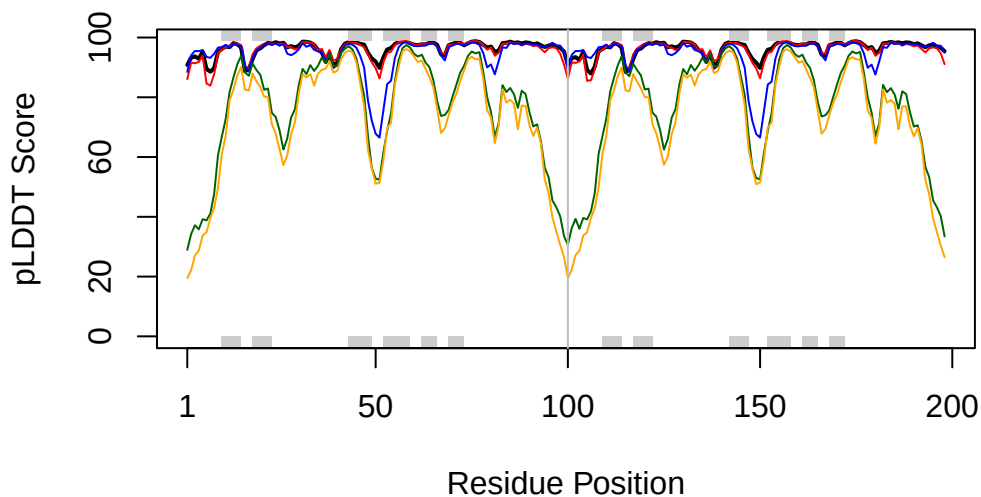


The RMSD heatmap compares similarity between the five AlphaFold models. The lower RMSD values indicate models that are more similar, while higher RMSD values indicate greater differences between the predicted models.

```
plotb3(pdbb$b[1,], typ = "l", lwd = 2, sse = pdb,
       ylab = "pLDDT Score",
       xlab = "Residue Position")

points(pdbb$b[2,], typ = "l", col = "red")
points(pdbb$b[3,], typ = "l", col = "blue")
points(pdbb$b[4,], typ = "l", col = "darkgreen")
points(pdbb$b[5,], typ = "l", col = "orange")

abline(v = 100, col = "gray")
```



This plot shows the pLDDT confidence scores across all five AlphaFold models. The higher PLDDT values indicate higher confidence in the predicted residue positions.

```
core <- core.find(pdbbs)
```

```
core size 197 of 198 vol = 9885.822
core size 196 of 198 vol = 6896.71
core size 195 of 198 vol = 1337.847
core size 194 of 198 vol = 1040.67
core size 193 of 198 vol = 951.857
core size 192 of 198 vol = 899.083
core size 191 of 198 vol = 834.732
core size 190 of 198 vol = 771.338
core size 189 of 198 vol = 733.065
core size 188 of 198 vol = 697.28
core size 187 of 198 vol = 659.742
core size 186 of 198 vol = 625.273
core size 185 of 198 vol = 589.541
core size 184 of 198 vol = 568.253
core size 183 of 198 vol = 545.015
core size 182 of 198 vol = 512.889
core size 181 of 198 vol = 490.723
core size 180 of 198 vol = 470.266
```

core size 179 of 198	vol = 450.731
core size 178 of 198	vol = 434.735
core size 177 of 198	vol = 420.337
core size 176 of 198	vol = 406.658
core size 175 of 198	vol = 393.334
core size 174 of 198	vol = 382.395
core size 173 of 198	vol = 372.858
core size 172 of 198	vol = 356.994
core size 171 of 198	vol = 346.567
core size 170 of 198	vol = 337.446
core size 169 of 198	vol = 326.659
core size 168 of 198	vol = 314.95
core size 167 of 198	vol = 304.127
core size 166 of 198	vol = 294.552
core size 165 of 198	vol = 285.648
core size 164 of 198	vol = 278.884
core size 163 of 198	vol = 266.765
core size 162 of 198	vol = 258.994
core size 161 of 198	vol = 247.723
core size 160 of 198	vol = 239.84
core size 159 of 198	vol = 234.963
core size 158 of 198	vol = 230.062
core size 157 of 198	vol = 221.985
core size 156 of 198	vol = 215.62
core size 155 of 198	vol = 206.793
core size 154 of 198	vol = 196.984
core size 153 of 198	vol = 188.539
core size 152 of 198	vol = 182.262
core size 151 of 198	vol = 176.954
core size 150 of 198	vol = 170.712
core size 149 of 198	vol = 166.119
core size 148 of 198	vol = 159.796
core size 147 of 198	vol = 153.767
core size 146 of 198	vol = 149.092
core size 145 of 198	vol = 143.657
core size 144 of 198	vol = 137.138
core size 143 of 198	vol = 132.517
core size 142 of 198	vol = 127.231
core size 141 of 198	vol = 121.574
core size 140 of 198	vol = 116.775
core size 139 of 198	vol = 112.57
core size 138 of 198	vol = 108.17
core size 137 of 198	vol = 105.133

core size 136 of 198	vol = 101.249
core size 135 of 198	vol = 97.374
core size 134 of 198	vol = 92.974
core size 133 of 198	vol = 88.184
core size 132 of 198	vol = 84.029
core size 131 of 198	vol = 81.898
core size 130 of 198	vol = 78.019
core size 129 of 198	vol = 75.272
core size 128 of 198	vol = 73.052
core size 127 of 198	vol = 70.695
core size 126 of 198	vol = 68.975
core size 125 of 198	vol = 66.694
core size 124 of 198	vol = 64.394
core size 123 of 198	vol = 62.092
core size 122 of 198	vol = 59.045
core size 121 of 198	vol = 56.629
core size 120 of 198	vol = 54.016
core size 119 of 198	vol = 51.806
core size 118 of 198	vol = 49.652
core size 117 of 198	vol = 48.193
core size 116 of 198	vol = 46.648
core size 115 of 198	vol = 44.752
core size 114 of 198	vol = 43.292
core size 113 of 198	vol = 41.093
core size 112 of 198	vol = 39.147
core size 111 of 198	vol = 36.472
core size 110 of 198	vol = 34.117
core size 109 of 198	vol = 31.47
core size 108 of 198	vol = 29.448
core size 107 of 198	vol = 27.325
core size 106 of 198	vol = 25.822
core size 105 of 198	vol = 24.15
core size 104 of 198	vol = 22.648
core size 103 of 198	vol = 21.069
core size 102 of 198	vol = 19.953
core size 101 of 198	vol = 18.3
core size 100 of 198	vol = 15.723
core size 99 of 198	vol = 14.841
core size 98 of 198	vol = 11.646
core size 97 of 198	vol = 9.434
core size 96 of 198	vol = 7.354
core size 95 of 198	vol = 6.179
core size 94 of 198	vol = 5.666

```

core size 93 of 198  vol = 4.705
core size 92 of 198  vol = 3.665
core size 91 of 198  vol = 2.77
core size 90 of 198  vol = 2.151
core size 89 of 198  vol = 1.715
core size 88 of 198  vol = 1.15
core size 87 of 198  vol = 0.874
core size 86 of 198  vol = 0.685
core size 85 of 198  vol = 0.528
core size 84 of 198  vol = 0.37
FINISHED: Min vol ( 0.5 ) reached

```

```
core.inds <- print(core, vol = 0.5)
```

```

# 85 positions (cumulative volume <= 0.5 Angstrom^3)
  start end length
1      9  49     41
2     52  95     44

```

```
x <- pdbfit(pdb, core.inds, outpath = "corefit_structures")
```

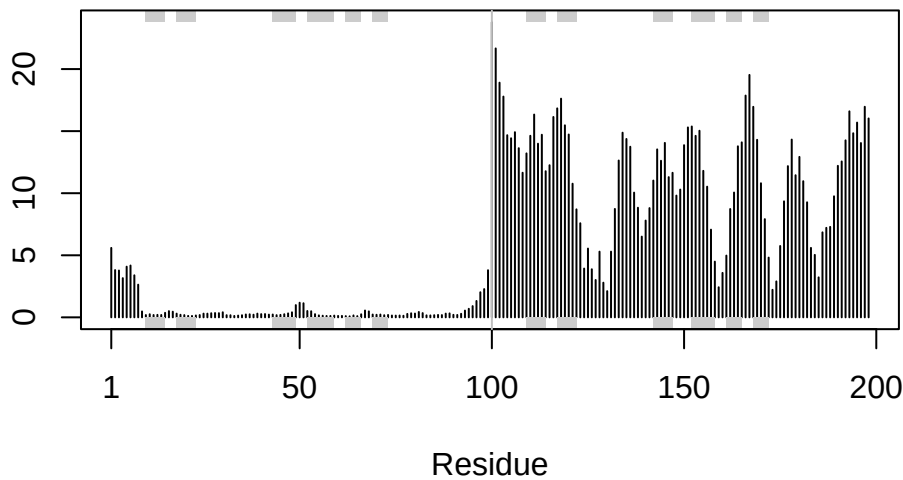
The `core.find()` function helps the structure consistent core regions shared across the five AlphaFold models. These core positions are used with `pdbfit()` to superpose the models more accurately.

```

rf <- rmsf(x)

# Plot rf
plotb3(rf, sse = pdb)
abline(v = 100, col = "gray", ylab="RMSF")

```

This RMSF plot shows how each residue position varies across the fitted AlphaFOLD models., a lower RMSF value indicates residues that are more structurally consistent, while higher values are more flexible.

```
results_dir <- "hivpr_dimer_23119/"
```

```
library(jsonlite)
```

```
pae_files <- list.files(path = results_dir,
                        pattern = ".*model.*\\.json",
                        full.names = TRUE)
```

```
pae_files
```

```
[1] "hivpr_dimer_23119//hivpr_dimer_23119_scores_rank_001_alphafold2_multimer_v3_model_4_see
[2] "hivpr_dimer_23119//hivpr_dimer_23119_scores_rank_002_alphafold2_multimer_v3_model_1_see
[3] "hivpr_dimer_23119//hivpr_dimer_23119_scores_rank_003_alphafold2_multimer_v3_model_5_see
[4] "hivpr_dimer_23119//hivpr_dimer_23119_scores_rank_004_alphafold2_multimer_v3_model_2_see
[5] "hivpr_dimer_23119//hivpr_dimer_23119_scores_rank_005_alphafold2_multimer_v3_model_3_see
```

```
pae1 <- read_json(pae_files[1], simplifyVector = TRUE)
pae5 <- read_json(pae_files[5], simplifyVector = TRUE)
```

```
attributes(pae1)
```

```
$names
```

```
[1] "plddt" "max_pae" "pae" "ptm" "iptm"
```

```
head(pae1$plddt)
```

```
[1] 90.81 93.25 93.69 92.88 95.25 89.44
```

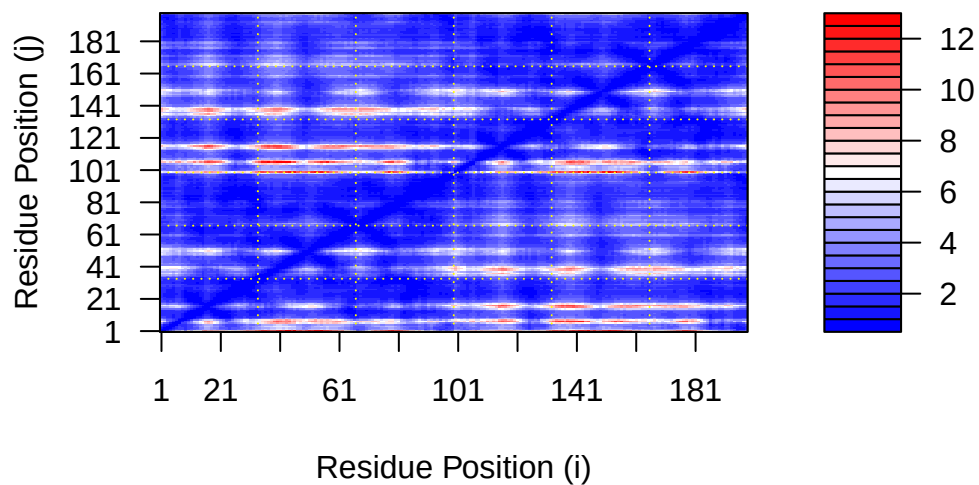
```
pae1$max_pae
```

```
[1] 12.84375
```

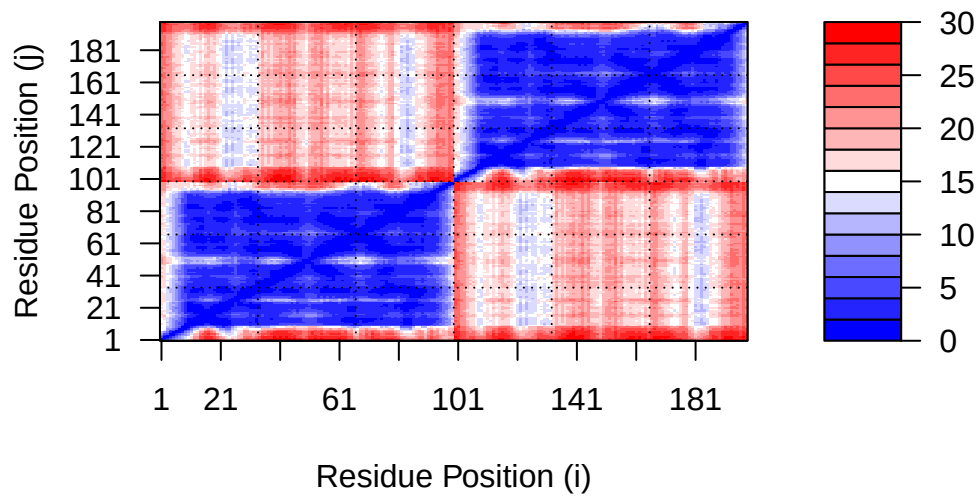
```
pae5$max_pae
```

```
[1] 29.59375
```

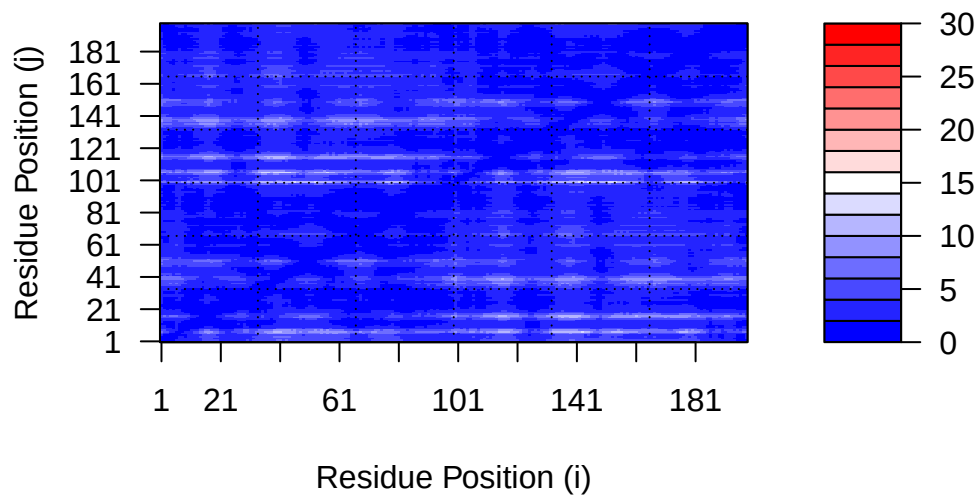
```
plot.dmat(pae1$pae,  
          xlab = "Residue Position (i)",  
          ylab = "Residue Position (j)")
```



```
plot.dmat(pae5$pae,
  xlab = "Residue Position (i)",
  ylab = "Residue Position (j)",
  grid.col = "black",
  zlim = c(0, 30))
```



```
plot.dmat(pae1$pae,
  xlab = "Residue Position (i)",
  ylab = "Residue Position (j)",
  grid.col = "black",
  zlim = c(0, 30))
```



```
pae2 <- read_json(pae_files[2], simplifyVector=TRUE)
pae2$max_pae
```

```
[1] 20.07812
```

```
pae3 <- read_json(pae_files[3], simplifyVector=TRUE)
pae3$max_pae
```

```
[1] 16.09375
```

```
pae4 <- read_json(pae_files[4], simplifyVector=TRUE)
pae4$max_pae
```

```
[1] 29.60938
```

The PAE plots show the predicted alignment error between the residue positions in the AlphaFold models, lower PAE indicate higher confidence, and higher values indicate more uncertainty.

```
aln_file <- list.files(path = results_dir,
                      pattern = ".a3m$",
                      full.names = TRUE)
aln_file
```

```
[1] "hivpr_dimer_23119//hivpr_dimer_23119.a3m"
```

```
aln <- read.fasta(aln_file[1], to.upper = TRUE)
```

```
[1] " ** Duplicated sequence id's: 101 **"
```

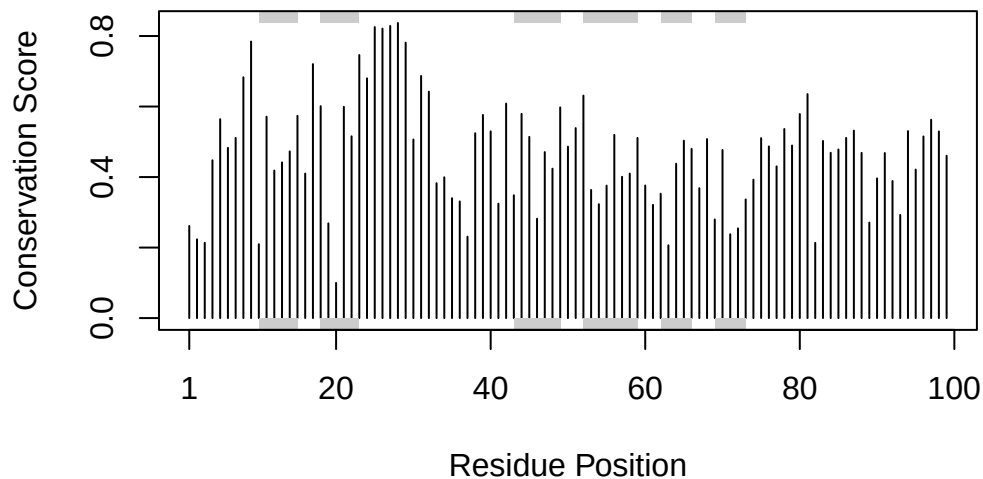
```
[2] " ** Duplicated sequence id's: 101 **"
```

```
dim(aln$ali)
```

```
[1] 5397 132
```

```
sim <- conserv(aln)
```

```
plotb3(sim[1:99],
       sse = trim.pdb(pdb, chain = "A"),
       ylab = "Conservation Score",
       xlab = "Residue Position")
```



This plot shows the residue conservation across sequence alignment.

```
con <- consensus(aln, cutoff = 0.9)
con$seq
```

```
[1] "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-"
[19] "-" "-" "-" "-" "-" "-" "D" "T" "G" "A" "-" "-" "-" "-" "-" "-" "-" "-"
[37] "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-"
[55] "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-"
[73] "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-"
[91] "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-"
[109] "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-"
[127] "-" "-" "-" "-" "-" "-"
```

```
m1.pdb <- read.pdb(pdbfiles[1])
occ <- vec2resno(c(sim[1:99], sim[1:99]), m1.pdb$atom$resno)
write.pdb(m1.pdb, o = occ, file = "m1_conserv.pdb")
```

This code maps the conservation scores onto the occupancy column of the ranked AlphaFold.